# Artificial Intelligence
# Lecture 0: Course overview

Prof. Dr. Pierre-Alexandre Murena

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# About the course

**Lecturer:** Pierre-Alexandre Murena (pierre-alexandre.murena@tuhh.de)

**Teaching assistant:** Niklas Dieckow

**Organization:**

- Lectures: Wednesdays 8:00-9:30 (A 0.13)
- Lab sessions: Fridays 8:00-9:30 (D 2.022) + Mondays 8:00-9:30 (A 0.10)

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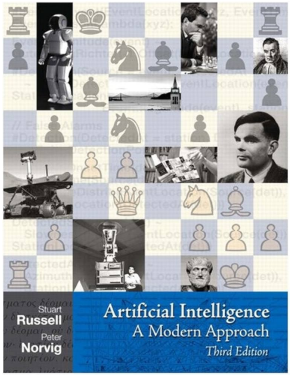
**Evaluation:** The final grade is computed as follows:

- Written exam
- Optional bonus questions (10% in total)
- Final grade: 50 % of the points to pass

# How to work during this course

- No slide will be available before the lecture
- Attendance to the lectures is not compulsory, but strongly encouraged
- Be active during the lectures: ask questions, participate...
- Take notes: the slides are **not** enough!
- Attendance to the lab sessions is strongly encouraged.
- If too many absences are observed, I will write the names of people being present. If 2 or more absences = malus on the final grade

# Reference book




The book PDF is available for free:
`https://people.engr.tamu.edu/guni/csce421/files/AI_Russell_Norvig.pdf`

# Motivation

**Why is AI needed when ML works so well?**

# Planning actions

PA **You**

I need to grab a banana hanging from a branch but I am a bit too short for that. Nearby, I can spot a big box, a lamp, an umbrella, a jar of peanut butter, some sun glasses, a PS5 and a woolen sweater. How should I proceed?

# Planning actions

**ChatGPT**

To reach the banana hanging from the branch, you can use the items nearby creatively. Here's a step-by-step plan:

1. **Use the Box as a Step Stool:**
   - Place the big box under the banana branch to elevate yourself.
2. **Climb onto the Box:**
   - Climb onto the box to get closer to the banana.
3. **Use the Lamp Cord as a Lasso:**
   - If the lamp has a cord, you can detach it and use it as a makeshift lasso to hook the banana and pull it closer.
4. **Secure the Banana with the Umbrella:**
   - Once you've lassoed the banana, use the umbrella to catch it and prevent it from falling.
5. **Use the Jar of Peanut Butter as Bait:**
   - If the banana is out of reach even with the lamp cord, spread some peanut butter on the umbrella as bait to lure the banana closer.
6. **Wear the Sunglasses for Protection:**
   - Wear the sunglasses to protect your eyes from any debris or branches while reaching for the banana.
7. **Use the PS5 as Additional Elevation (Optional):**
   - If the box isn't tall enough, you can stack the PS5 on top of it to gain extra height.
8. **Wrap the Woolen Sweater Around the Banana (Optional):**
   - If the banana is slippery or difficult to grip, wrap the woolen sweater around it to provide better traction.

By following these steps and using the items creatively, you should be able to reach the banana hanging from the branch!

# Reasoning

**You**

The language is based on declensions.
To formulate the accusative case, we add the suffix -to if the word ends up with a single vowel, -t if it ends with a double vowel, and -ato otherwise.
In addition, add re- prefix for the plural.

Give the singular and plural accusative forms of the following words:
- Adobe
- Windows
- Yahoo

**ChatGPT**

Sure, here are the singular and plural accusative forms of the given words based on the rules you provided:

1. **Adobe**
   - Singular Accusative: Adobeato (ends with a consonant, so add -ato)
   - Plural Accusative: Readobero (pluralized with "re-" prefix, ends with a single vowel, so add -to)
2. **Windows**
   - Singular Accusative: Windowsto (ends with a vowel, so add -to)
   - Plural Accusative: Rewindows (pluralized with "re-" prefix)
3. **Yahoo**
   - Singular Accusative: Yahoot (ends with a double vowel, so add -t)
   - Plural Accusative: Reyahoaht (pluralized with "re-" prefix, ends with a double vowel, so add -t)

These forms represent the accusative case of the nouns in both singular and plural, following the declension rules you provided.

# Reasoning (again)

**You**

How many phalanges is there in a group of 5 people (including one person with only one foot)

**ChatGPT**

To calculate the total number of phalanges in a group of 5 people, including one person with only one foot, we can use the following approach:

- Each person typically has 14 phalanges in total (3 in each finger and 2 in each thumb).
- Since one person has only one foot, they have a total of 5 phalanges (2 in the big toe and 3 in the remaining toes).

So, for 4 people with two feet each:
4 people * 2 feet/person * 5 phalanges/foot = 40 phalanges

Plus, for the one person with only one foot:
1 person * 1 foot/person * 5 phalanges/foot = 5 phalanges

Total number of phalanges = 40 phalanges (from 4 people with two feet each) + 5 phalanges (from 1 person with only one foot) = 45 phalanges.

Therefore, in a group of 5 people, including one person with only one foot, there are a total of 45 phalanges.

# Language

**You**

Let's play again. The first one who uses the word "yes" loses. Synonyms are allowed. Ready?

**ChatGPT**

Sure, I'm ready! Let's begin.

# Language

**You**

What is the converse of "no"?

**ChatGPT**

The converse of "no" is "yes." However, since we're playing a game where we avoid using the word "yes," I'll provide a synonym instead: "negative."

# Course outline

**Part 1: Problem solving**

- Search strategies in graphs and trees
- Search with partial observability
- Adversarial search
- Game theory

**Part 2: Knowledge representation and reasoning**

**Part 3: Toward modern AI**

# Course outline

**Part 1: Problem solving**

**Part 2: Knowledge representation and reasoning**

- Logic: Propositional logic, First-order logic
- Modal logic
- Description logics
- Ontologies and RDF

**Part 3: Toward modern AI**

# Course outline

**Part 1: Problem solving**

**Part 2: Knowledge representation and reasoning**

**Part 3: Toward modern AI**

- Natural language processing
- Probabilistic reasoning
- Neuro-symbolic AI